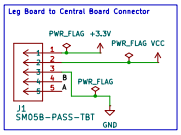
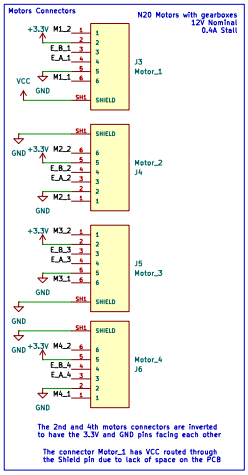
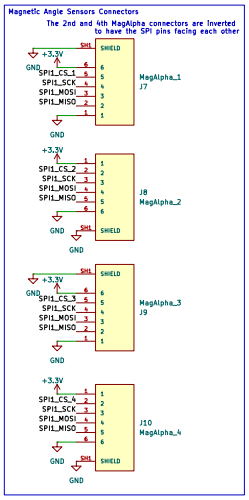
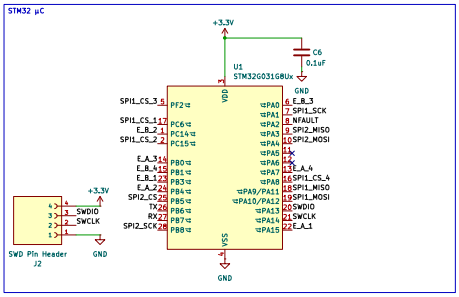
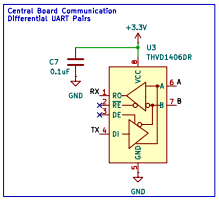

Motors Driver
U2
DRV8908QPWPRQ1
Motor Driver always active
=> NSLEEP tied to logic 1
C5 sizing : Practical TI method
Power x 4 <=> 12*1.6^4 = 78.8
Other capa values taken from
the TI DRV8906 datasheet
Contains overvoltage protection


Leg Board to Central Board Connector
PWR_FLAG +3.3V
PWR_FLAG VCC
PWR_FLAG
J1
SM05B-PASS-TBT


Motors Connectors
N20 Motors with gearboxes
12V Nominal
0.4A Stall
J3
Motor_1
Motor_2
J4
J5
Motor_3
Motor_4
J6
The 2nd and 4th motors connectors are inverted
to have the 3.3V and GND pins facing each other
The connector Motor_1 has VCC routed through
the Shield pin due to lack of space on the PCB


Magnetic Angle Sensors Connectors
The 2nd and 4th MagAlpha connectors are inverted
to have the SPI pins facing each other
MagAlpha_1
J7
J8
MagAlpha_2
MagAlpha_3
J9
J10
MagAlpha_4


STM32 µC
U1
STM32G031G8Ux
SWD Pin Header
J2


Central Board Communication
Differential UART Pairs
U3
THVD1406DR